

Pranav Kumar Soma

INFRASTRUCTURE • DISTRIBUTED SYSTEMS • PLATFORM ENGINEERING

pranavsoma11@gmail.com • linkedin.com/in/pranav-kumar-soma • pranavsoma.me

San Francisco Bay Area, CA • Available January 2027

Systems engineer with production performance work at enterprise scale (7s to sub-1s over millions of records, 10K+ calls/day), real-time C/C++ and ROS control pipelines, multi-threaded HPC over 5.5M-node graphs, and container and CI/CD ownership. Graduate coursework in operating systems, distributed systems, parallel computing, and security.

EDUCATION

M.S. Computer Science, University of California San Diego <i>GPA 4.00/4.00 • Jacobs School of Engineering • completed in 1 year</i>	Jan 2026 – Dec 2026 La Jolla, CA
B.S. Computer Science, University of California San Diego <i>GPA 3.96/4.00 • Cum Laude, CSE Department Honors, Provost Honors every quarter</i>	Sep 2023 – Dec 2025 La Jolla, CA
Coursework: Operating Systems, Distributed Systems, Parallel Computing, Computer Security, Graduate Algorithms, Theory of Computation, Principles of Programming Languages, Data Systems for ML, Software Engineering.	

EXPERIENCE

Software Engineer Intern, ServiceNow <i>AI Voice Agent Analytics</i>	Jun 2026 – Sep 2026 San Diego, CA
- Cut report load 7s to under 1s over millions of records (up to 100 interaction records per call) through lazy loading, caching, and an iterative scalability-driven redesign of the query path across 20+ joined tables - Shipped the platform to production across 24+ enterprise customer instances serving 10K+ voice calls/day; built Playwright integration suites running in Jenkins CI/CD plus instance-level verification	

Founding Software Engineer, Stealth Startup <i>Autonomous systems • part-time</i>	Apr 2026 – Present
- Building the core operating system and software infrastructure from the ground up: hardware-agnostic, fault-tolerant, horizontally scalable across distributed edge nodes - Owning distributed messaging and networking for edge nodes on degraded links, low-latency C2 telemetry dashboards, and the simulation and hardware-in-the-loop harness the team tests real systems against	

Computer Vision & Embedded Systems Engineer, Triton Robotics <i>50-person competitive robotics team, CV and Embedded subteams</i>	Sep 2023 – Jun 2024 San Diego, CA
- Built real-time motor-control pipelines in C and C++ over ROS, compiling Python planning instructions into motor interactions across the control stack; designed low-level data pipelines and self-authored libraries - Implemented conditional pure-pursuit path planning, PID controllers, acceleration tuners, and contour-detection tracking; trained PyTorch vision models for autonomous target acquisition	

Honors & M.S. Thesis Researcher, UCSD CSE, Computational Redistricting	Sep 2024 – Present
- Built multi-threaded graph-partitioning generators over Census 2020 data at 158K to 5.5M blocks/state, parallelized across San Diego Supercomputer Center nodes over a 5 to 18M-row Parquet store; pymetis, GeoPandas, Shapely - Implemented a Sequential Monte Carlo sampler (spanning-tree balanced cuts, N=128 particles), greedy DP with Kernighan–Lin refinement, and XGBoost-guided partitioning; reproduces official district counts for AL, CA, TX	

AI Research Fellow & Agentic Systems Lead, Laboratory for Emerging Intelligence, UCSD	Nov 2024 – Present
Lead 6 engineers on a Teradata-funded agentic platform (100+ pilot users): 60+ MCP servers fronting all 600 UCSD APIs, a deterministic fault-graceful workflow executor, and in-unit memory management; GCP Cloud Run, FastAPI, MongoDB	

Software Engineer Intern, ServiceNow <i>AI Voice Agents • Technical Skills Award, top intern of 800+</i>	Jun 2025 – Sep 2025 San Diego, CA
Owned the AWS Lex V1 to V2 platform migration end to end (8 intents, 48+ slots) ahead of end-of-life: 45% lower agent-response and tool-output latency, built on Lambda and REST APIs over ServiceNow tables. Return offer	

Software Developer, Triton Software Engineering <i>Engineering Manager, Sep 2024 – Jun 2025</i>	Sep 2023 – Jun 2025 San Diego, CA
Co-built TSE Constellation, the org's published React/TypeScript design system (@tritonse/tse-constellation; Vite, Storybook, atomic design) adopted org-wide by 100+ engineers; led 11 developers as Engineering Manager	

LEADERSHIP & HONORS

Co-Founder & President, Real World AI Network • **7,000+ student** federation • **President**, CSE Society @ UCSD • **300+** members, **\$50K+** raised, first **Linux Foundation** hackathon.

Honors: ServiceNow Technical Skills Award • Best Project SDTC 2025 • Cum Laude & CSE Department Honors • National Merit Finalist • USACO Silver.

TECHNICAL SKILLS

Systems & Infrastructure: Docker, Kubernetes, Jenkins CI/CD, GCP (Cloud Run), AWS (Lambda, Lex), Git, Linux, ROS; distributed, parallel, and real-time systems, multi-threaded and multi-node computing, system design, finite state machines, embedded systems, PID control, hardware-in-the-loop

Languages: C, C++, Python, Java, TypeScript, JavaScript, SQL, Bash, ARM Assembly

Data & Application: PostgreSQL, MongoDB (vector), MySQL, Parquet/PyArrow, NumPy, Pandas, GeoPandas, pymetis; FastAPI, Flask, Node.js, Express, React, REST APIs; Playwright, Postman, Agile/Scrum; PyTorch, XGBoost, OpenCV, LLM agents and MCP